

THOMAS REESE

Full-Stack Software Engineer · AI Systems & Developer Tooling

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SUMMARY

Built and shipped **Codebase Intelligence** at Envoy: an 87,500-chunk **RAG** index over 22 repos via a Python **MCP server** on **AWS Bedrock**, hitting **Hit@1 0.785** at **~\$0.03/month** on a 200-query human-labeled eval. Four years across production Ruby on Rails, Ember.js, React 19, and Python. Daily work: **Pulumi/Terraform** IaC, RAG retrieval, **LLM evals**, and tool-use guardrails. Side project: **TraderDan**, a 14-agent live prop-firm trading system with 5,000+ tests.

PROFESSIONAL EXPERIENCE

Envoy | Full-Stack Software Engineer, Visitor Management San Francisco, CA | Nov 2024 – Present

- Shipped **Codebase Intelligence** at **Hit@1 0.785** / **~\$0.03/month**: a **RAG** platform indexing 22 internal repos (~87,500 AST-aware code chunks, tree-sitter parsed) on **Amazon S3 Vectors** and **Bedrock Knowledge Bases (Titan Text Embeddings V2)**, with a Python **MCP server** exposing six retrieval tools to Anthropic **Claude**. Retired per-engineer 6–8 GB local vector DBs across the eng org.
- Migrated the visitor-management dashboard from **Ember/Glimmer** to **React 19** across six surfaces (profile, printers, screeners, visits, matching-rules, documents) using **Module Federation 2.0**, **TanStack Query v5**, and **Radix UI**, shipping incrementally behind feature flags.
- Designed **AgentLock**, a phase-based **guardrails** and permission model for the 14 **agentic** AI tools embedded in the Visitors product — capability gating per request lifecycle, observability hooks, and policy enforcement at the MCP boundary.
- Consolidated duplicate visitor profiles across Envoy's customer organizations by shipping **Visitor Profile Merge** and **Block List V2** end-to-end over two quarters — Rails APIs, React merge UI, and the identity-matching lifecycle (expiration, disable toggle, search, deletion). Validated with TDD: **RSpec** on Rails, **Vitest + MSW** on React, plus staging-pod-exec mutation testing.
- Replaced fragile SKU-name string matching with a **ChargeBee feature-code subscription gate** on the Visitors Basic plan: counter table, gate-state API, 75% / 100% threshold mailer, and reset cron, spanning Rails and Ember in one sprint.
- Authored Envoy's internal AI playbook: AI-tooling onboarding guide, a six-layer agent memory reference, and a coding-agent **observability** writeup adopted by Envoy's Engineering Innovation Program.
- Sustained **5+ merged PRs/week** across nine repos on a stacked-PR (**Graphite**) and worktree workflow, spanning Rails, Ember (Octane / Glimmer / GTS), React 19, **Cerbos** authorization, and Pulumi / Terraform IAM.

Sprout Social | Front-End Engineer

Chicago, IL | May 2022 – Apr 2024

- Migrated core front-end components from **Flow** to **TypeScript**, eliminating a class of runtime type errors and unblocking strict-mode adoption across the codebase.
- Cut per-feature build time ~3 days across product teams by building and maintaining the **Seeds** shared React component library, standardizing UI primitives org-wide.
- Cut analyst handle time ~30% per batch by shipping a React UI for **bulk sentiment reclassification** (1,000 messages/batch) with **Redux** and virtualized lists.
- Built a **Query Builder** in React + TypeScript backed by **OpenAI GPT** keyword suggestions with debounced search, lowering query-setup time for social-listening analysts.
- Lifted listening-dashboard session engagement ~30% by launching a **faceted message-exploration tool** (React Context, Redux, React Router) with virtualized lists.

The Walt Disney Company | Software Engineering Intern

Remote | May 2021 – Nov 2021

- Cut ESPN editorial banner-creation time ~35% by building **Banner-Maker** — reusable React components with Redux state and team / player API integration.
- Cut form-validation errors ~60% by owning the React front-end for a **Student Loan Refinance** product — responsive UI, client-side validation, and back-end API integration.
- Raised automated a11y coverage ~70% by driving **WCAG** accessibility work with **Applitools** — screen-reader audits and semantic-HTML refactors.

Credit Karma | Full-Stack Web Developer Intern

San Francisco, CA | Jan 2020 – Dec 2020

- Built internal **loan-refinancing tools** across **React**, **Node.js**, **Ruby**, and **Python** services on **AWS**, and consolidated team **QA notes and tips** into a single static knowledge site that surfaced institutional best practices and unlocked faster onboarding for new teammates.

SKILLS

<i>AI & LLM:</i>	Model Context Protocol (MCP), Anthropic Claude SDK, OpenAI SDK, Gemini Flash, Amazon Bedrock, S3 Vectors, Titan Text Embeddings V2, RAG, agentic tool use, LLM evals, prompt engineering, function calling, AI guardrails, tree-sitter AST chunking, LangChain, LangGraph
<i>Vector / Data:</i>	ChromaDB, Pinecone, pgvector, PostgreSQL, Redis, Kafka, Sidekiq, MongoDB, Optuna
<i>Cloud & Infra:</i>	AWS (Bedrock, S3, IAM), Pulumi, Terraform, Kubernetes, Docker, OpenTelemetry, Datadog, Vercel
<i>Languages:</i>	TypeScript, JavaScript (ES6+), Python, Ruby, HTML5, CSS3, SQL, Flow (legacy)
<i>Frameworks & UI:</i>	React 19, Next.js, Ember.js (Octane / Glimmer / GTS), Ruby on Rails, FastAPI, Node.js, Express, Tailwind, Radix UI, SCSS, Luxon
<i>Build & State:</i>	Module Federation 2.0, Vite, Webpack 5, TanStack Query v5, Redux Toolkit, Axios, MSW v2, GraphQL, REST
<i>Testing:</i>	Vitest, RSpec, Playwright, Jest, Cypress, MSW v2, mutation testing
<i>Workflow & Sec:</i>	Graphite (stacked PRs), GitHub Actions, CircleCI, Cerbos, OAuth, JWT, WCAG accessibility

EDUCATION

Dominican University

B.S. Computer Science

River Forest, IL

Aug 2018 – May 2022

PROJECTS

TraderDan — Automated Trading

Python · Next.js · FastAPI · Anthropic Claude · TradeLocker · PostgreSQL · Optuna · ChromaDB

Live multi-asset trading system across five active prop-firm accounts (XAUUSD H1/M15 via TradeLocker).

Architecture: backtest engine, FastAPI webhook execution server with prop-firm risk gates (daily-loss, drawdown, ATR-spike, news blackout, weekend closer), a 14-agent AI team led by an OpenClaw orchestrator (Anthropic Claude tool use, Gemini Flash routing), an eval harness with drift detection, ChromaDB semantic memory, and a Next.js 15 dashboard. 5,000+ tests, stacked-PR workflow.

Blue Ridge Canine Academy [\[link\]](#)

React · Node.js · Express · MongoDB · Acuity Scheduling · Figma

Freelance build (\$800) for a local dog trainer: a full-stack scheduling and payments platform. React frontend, Node.js / Express backend, MongoDB persistence. Wireframed in Figma and hosted on GoDaddy.

HONORS & AWARDS

- **1st place**, Google × Code Nation Hackathon.
- **Won 24-hour Code for Good Hackathon** (JP Morgan), leading a front-end team shipping a site for the Girls in the Game non-profit.
- **2nd place**, ACCA Programming Contest — highest Dominican University finish to date. Completed **MLT Career Prep**; STEM mentor at **Code Nation**.